

GAMEVERSE

A themed, single-page web portal that opens into three interactive game-world showcases. Each world is a long-scroll experience whose palette, typography, particle weather, ambient audio and motion change together when the visitor navigates.

4

ROUTES

69 / 71

CHECKS PASSED

≈205 kB

GZIPPED JS + CSS FOR
THE PORTAL

0

CONSOLE ERRORS ON
ANY ROUTE

Gopi Patel
Project owner

React 19 · Vite 6
Stack

7–11 Aug 2026
Commit history

15 Sep 2026
Report date

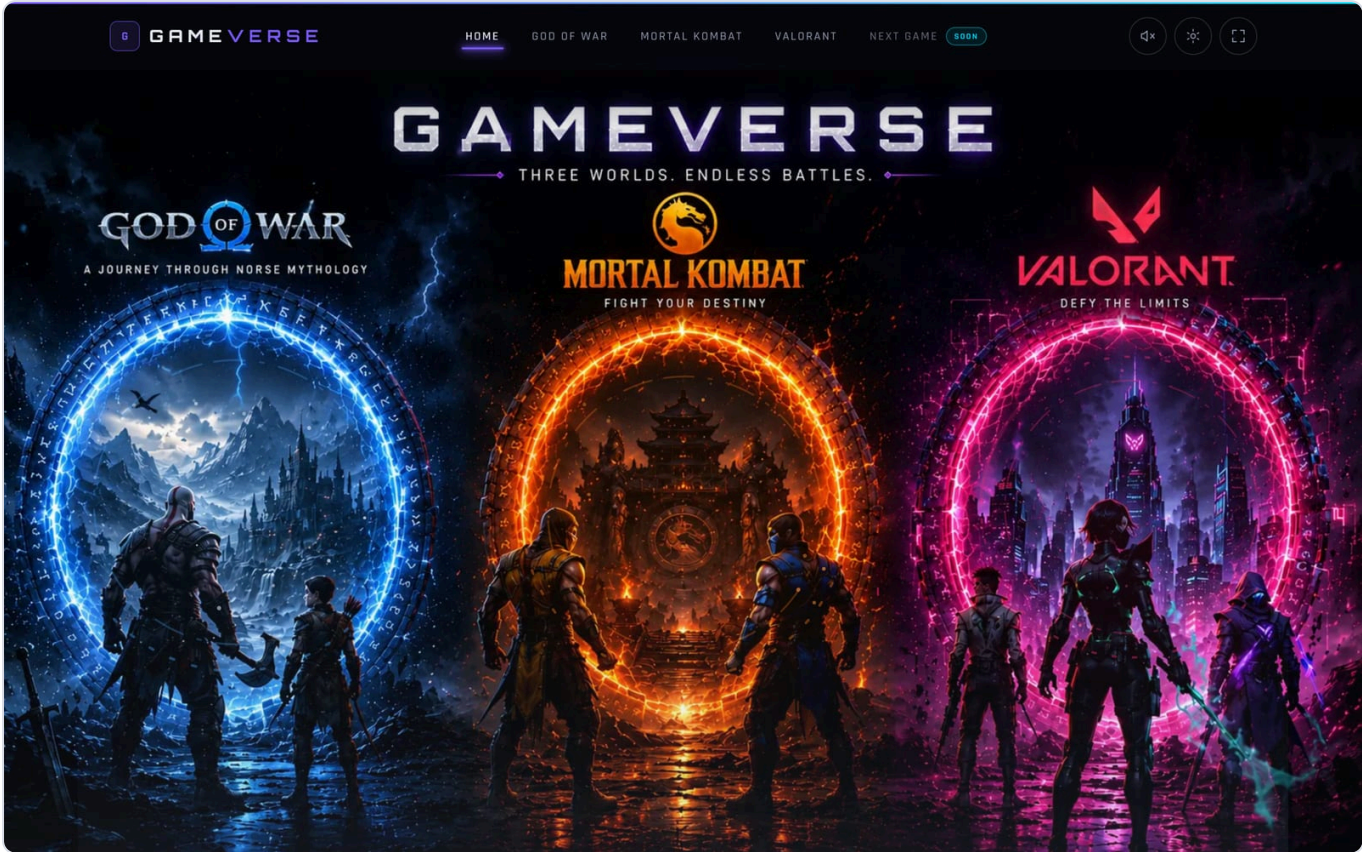
Report structure

01	Project Overview	What it is, status at a glance
02	Objectives & Requirements	What the project set out to do, with evidence
03	Solution Overview	How the requirements were met
04	Key Features & Capabilities	Status of every feature, with screenshots
05	System Architecture	Layers, data flow, cross-cutting systems
06	Technology Stack	What each dependency is used for
07	Core Workflows	Navigation, asset pipeline, media audio
08	Important Engineering Decisions	Problem → Decision → Implementation → Result
09	AI-Assisted Development	How AI was used, based on evidence
10	Implementation Details	Code excerpts from the key mechanisms
11	Challenges & Solutions	Problems recorded in commits and code
12	Testing & Verification	71 scripted checks on the production build
13	Security, Reliability & Data Integrity	Attack surface, fallbacks, asset integrity
14	Deployment & Usage	Commands, hosting, configuration
15	My Role & Contribution	Responsibilities supported by evidence
16	Results & Verified Metrics	Build, page weight, layout stability
17	Limitations	Defects, dormant code, known gaps
18	Future Improvements	Prioritised by impact
19	Conclusion	
20	Master Prompt	Rebuilding the project from scratch

Sources and method. The codebase at `E:\GAMEVERSE\GAMEVERSE` is the authority for every claim. Git history, the README, source media folders and configuration were reviewed alongside the code. All results in sections 12 and 16 were produced for this report on 15 September 2026: a production build, a 71-check scripted verification in headless Chrome against `vite preview`, network measurements, and a re-run of the asset pipeline. Where the README or the previous report disagreed with the code, the code was followed. Corrections include: the gallery is a uniform grid rather than a masonry layout, and `node_modules` and `dist/` are git-ignored rather than committed.

Project Overview

GAMEVERSE is a client-side React application with a cinematic landing page showing three portals. Each portal routes to a long-scroll page presenting one game world: **God of War**, **Mortal Kombat** or **Valorant**. The README describes it as a fan-made concept not affiliated with any studio, and the site footer says the same.



The portal (/). A supplied key-art plate. The application adds per-gate particle weather, gate glow, storm and fog layers, cursor parallax, and a labelled hotspot over each painted gate. Captured from the production build.

What it does	Status at a glance												
- Routes between a portal and three world pages without reloading. - Switches the visual identity with a single <code>data-theme</code> attribute: palette, display typeface, grain, particles, transition accent and ambient drone. - Renders each world from configuration, including section order. - Serves optimised media from a generated manifest. - Supports reduced motion, touch devices and a dedicated phone layout.	<table><tr><td>Core experience (4 routes)</td><td>WORKING</td></tr><tr><td>Scripted verification</td><td>69 / 71 PASS</td></tr><tr><td>Valorant agent captions</td><td>DEFECT</td></tr><tr><td>Boss section, signal-noise layer</td><td>DORMANT</td></tr><tr><td>Automated tests, linting, CI</td><td>NOT PRESENT</td></tr><tr><td>Deployment target</td><td>Vercel (configured)</td></tr></table>	Core experience (4 routes)	WORKING	Scripted verification	69 / 71 PASS	Valorant agent captions	DEFECT	Boss section, signal-noise layer	DORMANT	Automated tests, linting, CI	NOT PRESENT	Deployment target	Vercel (configured)
Core experience (4 routes)	WORKING												
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Deployment target	Vercel (configured)												

Scope in numbers

MEASURE

VALUE SOURCE

Source files in src/ (JS, JSX, CSS)	44 files · 5,837 lines	Line count
Build scripts	2 files · 228 lines	scripts/
Characters presented	34 (5 · 14 · 15)	games.js
Gallery items	51 (19 · 18 · 14)	games.js + manifest
Media referenced by the manifest	106 images · 12 videos	manifest.json
Optimised assets on disk	125 MB (31 MB WebP · 95 MB MP4)	public/assets
Source media (git-ignored)	339 MB	photos/

Objectives & Requirements

No formal specification exists in the repository. The objectives below come from the README, the organisation of the supplied media, and the constraints recorded in code comments and commits. Each is listed with the evidence for it.

Objectives

1. **One entry point, several distinct worlds.** Visitors should choose a world from a single portal and feel the whole site change around them.
2. **Presentation should follow the supplied art direction.** Section-level design plates and media were provided; the site had to use them faithfully instead of overwriting them.
3. **Adding a world should not require new page code.**
4. **Heavy media should stay usable** on ordinary connections and on phones.
5. **Motion and sound should never be forced** on visitors.
6. **The site should deploy as static files** with no image processing at build time.

Requirements traced to evidence

REQUIREMENT	EVIDENCE IN THE PROJECT
Portal plus three world pages	README; routes in <code>App.jsx</code> ; <code>photos/GOW</code> , <code>photos/mortal</code> , <code>photos/VALO</code>
Per-section layouts based on supplied plates	Source files named <i>hero section</i> , <i>brutality section</i> , <i>game mode section</i> , <i>story nd features</i> , <i>background for hero section</i> . A <code>games.js</code> comment notes one supplied plate "is a whole page mockup" whose baked-in UI was cropped off.
Different section order per world	A per-world <code>sections</code> array; comment: "the three worlds tell their stories in different orders"
Themed identity per world	Four <code>[data-theme]</code> token sets in <code>index.css</code>
Honest calls to action	Storefront lists are chosen per world; comments explain why God of War has no Xbox link and why Valorant links to Riot's client
Usable on phones	Commits on 10–11 Aug add a mobile nav sheet, mobile hero geometry and a portrait landing plate (<i>home page for mobile.png</i>)
No sound without consent	<code>SoundContext</code> comment: "when it is off nothing may make noise"; the hero comment calls a page that makes noise unprompted "hostile"
Reduced-motion support	CSS media query, <code>useReducedMotion</code> , and an in-app Motion switch
Static hosting without build-time image work	<code>vercel.json</code> ; README; <code>sharp</code> is a devDependency only
Non-affiliated fan concept	README notes; footer disclaimer; original geometric platform glyphs

Intended audience

Visitors browsing the three worlds on desktop or phone. The repository names no commercial client, and the project is not presented here as a commercial product.

Solution Overview

The solution rests on three mechanisms. Most of the rest of the code is presentation built on top of them.

1 · The route selects the theme

On every navigation the router writes `data-theme` on `<html>`. Every colour, the display typeface and the grain strength resolve from CSS variables scoped to that attribute, so no component branches on the active world for styling.

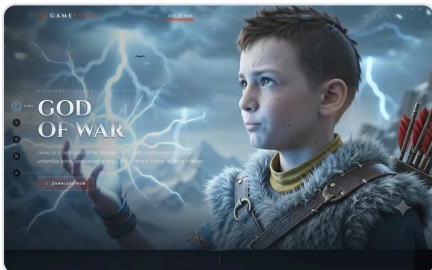
2 · Configuration drives the page

`games.js` holds all content and each world's section order. `GamePage` is a registry that maps section keys to components; it contains no layout of its own.

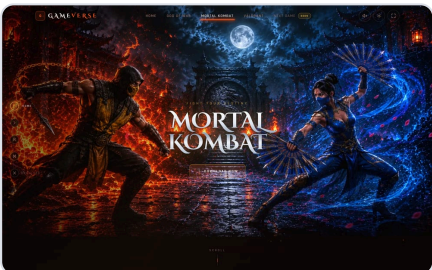
3 · A manifest provides the media

A build script converts `photos/` into WebP files and a JSON manifest. Components request assets by group and index or by name, and never hard-code a path.

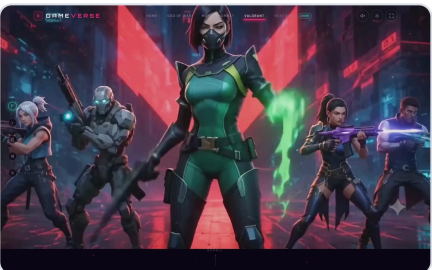
The three worlds



God of War · Cinzel, snow. Looping hero clip with fog, storm, torchlight and raven layers.



Mortal Kombat · Cinzel Decorative, embers. Copy is centred between the fighters.



Valorant · Orbitron, neon. An art-only plate over a looping clip, with a screen-reader heading.

WORLD	ROUTE	ACCENT	PARTICLES	SECTION ORDER
Portal	/	#7c5cff	dust	Key-art hero only
God of War	/god-of-war	#c8442f	snow	hero · characters · story · features · trailer · gallery · requirements · download
Mortal Kombat	/mortal-kombat	#ff5a1f	ember	same as God of War
Valorant	/valorant	#ff2d78	neon	hero · characters · story · maps · features · trailer · gallery · requirements · download

All four themes and all three section orders were confirmed in the rendered DOM during verification (section 12).

Key Features & Capabilities

Every feature below was checked against the running production build. **COMPLETE** means implemented and observed working. **DEFECT** means it works but has a verified fault. **DORMANT** means the code exists but never renders.

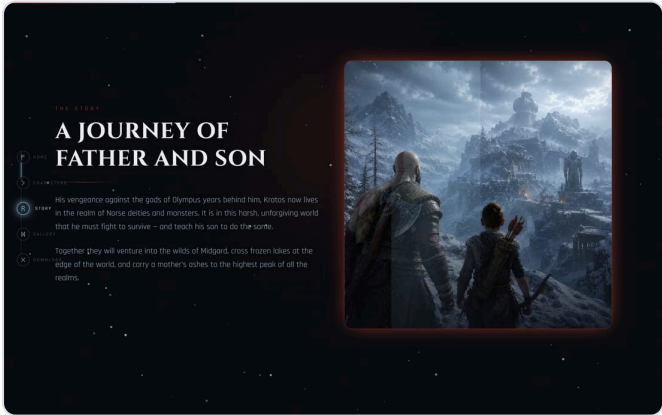
FEATURE	BEHAVIOUR	STATUS
Portal (desktop)	Key-art plate fitted below the navbar. Three gate hotspots positioned as fractions of the plate, each with its own particle canvas that animates only while visible; spinning rings, storm and fog layers, spring parallax, and a focus pull on hover.	COMPLETE
Portal (phone)	A dedicated portrait plate with the gates stacked, and three full-width tap bands.	COMPLETE
Per-route theming	Four token sets; 0.9 s colour transition; route shutter in the destination accent.	COMPLETE
Configured section order	Registry plus a per-world <code>sections</code> array; an unknown key logs a warning in development and renders nothing.	COMPLETE
Heroes	Video or still plate; GSAP mouse parallax on depth layers; a slow 26-second zoom. On the art-only hero, the single call to action appears during the clip's last 3 seconds, or immediately if the video fails.	COMPLETE
Valorant signal-noise layer	Scanline, glitch-tear and energy-wash overlay. It is gated behind <code>!artOnly</code> , and the Valorant hero is art-only.	DORMANT
Character rosters	Snap-scrolling rail with pointer tilt up to $\pm 5^\circ$. God of War cards play muted clips, with audio on hover when sound is enabled.	COMPLETE
Valorant caption suppression	The config sets <code>hideArtLabels</code> , but <code>Characters</code> reads <code>hideAgentLabels</code> . All 15 cards print captions over names already in the artwork.	DEFECT
Story	Scroll-linked parallax ($\pm 8\%$, scale $1.15 \rightarrow 1 \rightarrow 1.15$) with a moving light band; panel can follow the artwork's aspect ratio.	COMPLETE
Features	Five cards; GSAP ScrollTrigger stagger; pointer tilt up to $\pm 3^\circ$.	COMPLETE
Maps (Valorant)	Panel sized to the plate's aspect ratio, with a holographic grid and scan sweep on hover. One plate exists, so three of the four configured map names are unused.	COMPLETE
Trailer	Poster facade; the YouTube embed (<code>youtube-nocookie.com</code>) is created only on click.	COMPLETE
Gallery & lightbox	Uniform 2/3/4-column grid of 4:3 tiles. Lightbox with Escape and wrapping arrow keys; video items play with controls.	COMPLETE
Requirements & download	Minimum and recommended spec cards; per-world storefront links in new tabs with <code>noopener noreferrer</code> .	COMPLETE
Journey rail	Fixed rune rail at ≥ 1280 px. Scroll-spy uses IntersectionObserver and lists only the sections present, in document order.	COMPLETE
Chrome	Loading screen (3.2 s failsafe), page transition, custom cursor and mouse glow (fine pointers only), scroll progress bar, retracting nav.	COMPLETE
Settings	Ambience, Motion and Fullscreen switches; on phones, Motion and Fullscreen move into the menu sheet.	COMPLETE

Ambient audio	Procedural Web Audio drone per theme. State changes were verified; audible output was not measured.	COMPLETE
Reduced motion	OS preference and in-app switch; Lenis and the cursor are not started when motion is reduced.	COMPLETE
SEO	Per-route title, description, canonical, Open Graph and Twitter tags; sitemap and robots.txt. Client-side only.	COMPLETE
Boss encounters	218-line section with auras and power meters. No world lists <code>bosses</code> .	DORMANT
Stats band, trailer strip	Unreferenced; the landing comment says they were dropped. The stats band contains placeholder figures such as "2M+ players".	UNUSED
"Next Game" nav slot	An inert teaser with no route, by design.	PLACEHOLDER

Feature walkthrough



Character rail (God of War). The journey rail at the left edge lists only the sections this page contains.



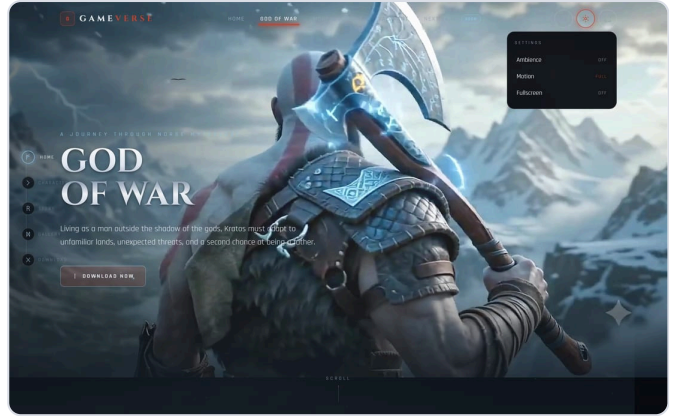
Story. Parallax panel pinned to the artwork's 1:1 aspect ratio.



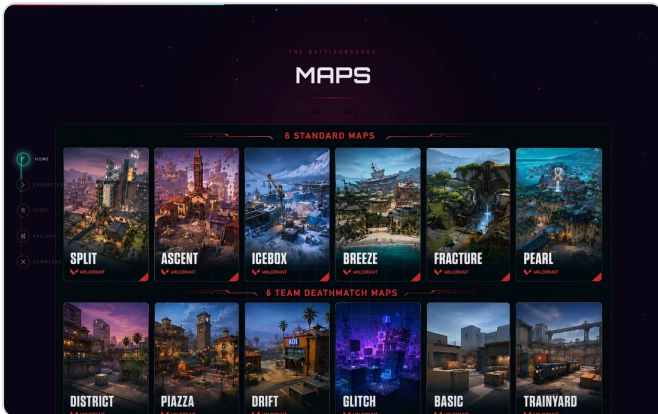
Story (Mortal Kombat). Pinned to 16:10 so both fighters are shown whole.



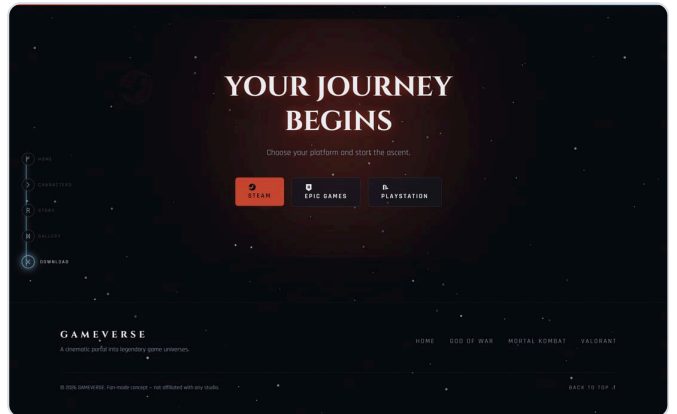
Lightbox. Keyboard navigation verified: Escape closes it; arrow keys advance and wrap.



Settings. Ambience, Motion and Fullscreen, shown over the God of War hero clip.



Maps (Valorant). A single plate at its own aspect ratio; captions hidden because the art is already labelled.

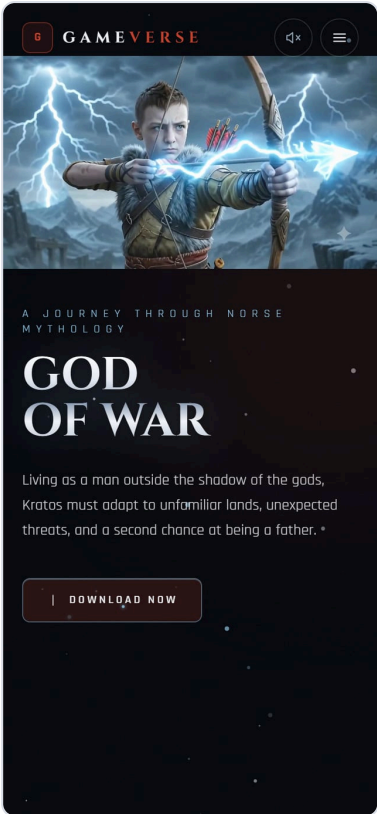


Download. Storefront links chosen per world, followed by the footer disclaimer.

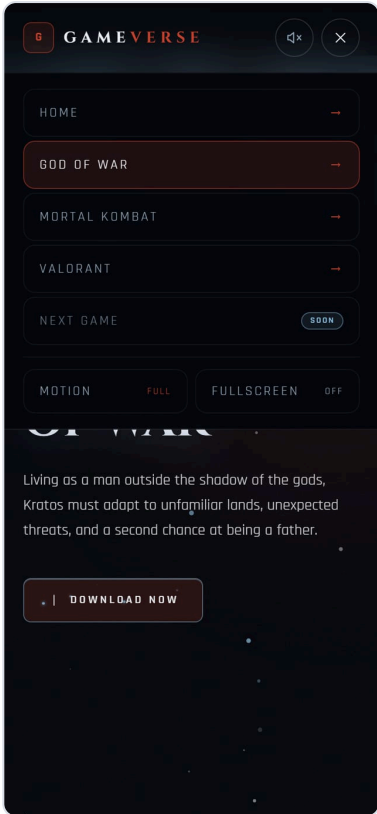
Phone layout (390 × 844, touch emulation)



Portal. Portrait plate; each gate is a full-width tap band.



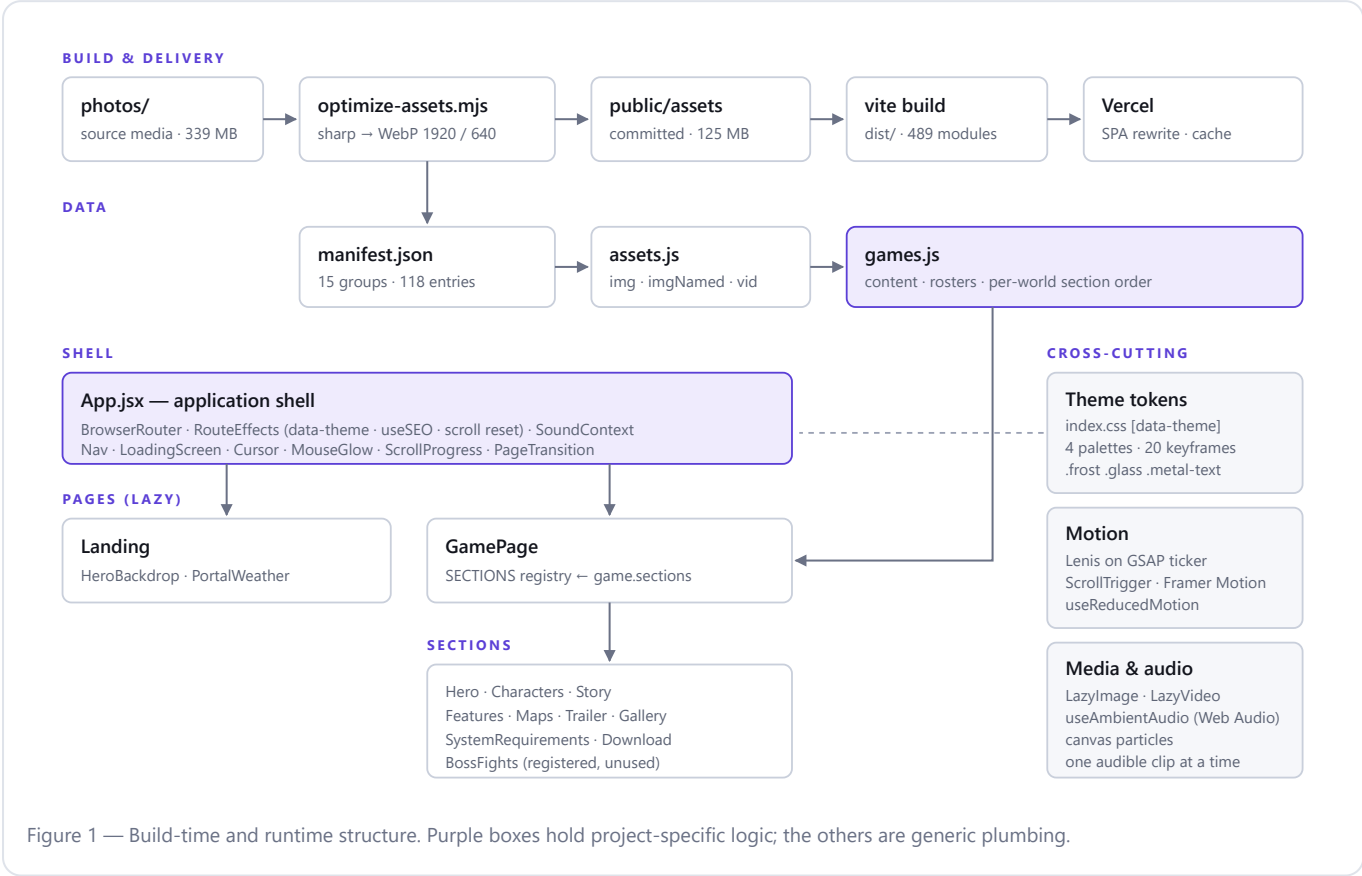
Hero. The art becomes a banner with the copy stacked beneath it.



Menu sheet. Motion and Fullscreen move here from the header.

System Architecture

A static single-page application with no backend. Build-time tooling produces media and a manifest; at runtime, a data layer feeds a shell, two lazily loaded pages and a set of generic sections. Theme, motion and media/audio cut across every layer.



Layer responsibilities

LAYER	MODULES	RESPONSIBILITY
Build	scripts/optimize-assets.mjs, contact-sheet.mjs	Convert and name media, emit the manifest; identify unlabelled images for development.
Data	manifest.json, assets.js, games.js	Typed, fault-tolerant asset lookup; all content and per-world layout configuration.
Shell	App.jsx, components/*	Routing, theme attribute, SEO, scroll reset, persistent chrome, sound state.
Pages	pages/Landing, pages/GamePage	Lazy route entry points; GamePage maps configuration to sections.
Sections	sections/*	Generic over a <code>game</code> prop; no world-specific branching except scoped hero atmosphere.
Hooks & context	useSmoothScroll, useAmbientAudio, useReducedMotion, useSEO, SoundContext	Shared behaviour with a single owner for each concern.

State management is local component state plus one context (`SoundContext`). There is no global store, backend, API or database.

Technology Stack

TECHNOLOGY	VERSION	USED FOR	WHY IT FITS
React	^19.1.0	UI runtime (Strict Mode)	Component model for a config-driven section registry
React Router	^7.6.2	Client routing	The pathname also selects the theme
Vite	^6.3.5 (6.4.3 resolved)	Dev server, build, preview	Route-level code splitting and manual chunking
Tailwind CSS	^4.1.10	Utility styling via <code>@tailwindcss/vite</code>	Utilities read theme variables directly, e.g. <code>bg-[var(--accent)]</code>
Framer Motion	^12.18.1	Page transitions, scroll-linked parallax, reveals, springs, presence	Declarative animation tied to React state
GSAP + ScrollTrigger	^3.13.0	Feature stagger, hero parallax and zoom, ticker for Lenis	A single RAF loop shared with smooth scrolling
Lenis	^1.3.4	Smooth scrolling	Can be driven by an external ticker
sharp (dev only)	^0.35.3	WebP conversion, contact sheets	Kept out of the production build
Web APIs	—	Web Audio, IntersectionObserver, ResizeObserver, MutationObserver, Canvas 2D, Fullscreen, User Activation	Audio synthesis and viewport-aware media without extra dependencies
Google Fonts	—	Cinzel, Cinzel Decorative, Orbitron, Anton, Rajdhani	Each theme selects a face through <code>--font-display</code>
Vercel	—	Static hosting (configured, not verified live)	SPA rewrite and immutable asset caching in <code>vercel.json</code>

Deliberately not used

No TypeScript, no state library, no UI component kit, no CSS-in-JS, no icon library (icons and runes are inline SVG paths), no audio files, and no analytics.

Code split in the production build

CHUNK	RAW	GZIP	LOADED
index	283.04 kB	91.81 kB	Always (shell)
motion	227.53 kB	78.43 kB	Always (framer-motion, gsap, lenis)
vendor	49.14 kB	17.36 kB	Always (react, react-dom, router)
GamePage	49.34 kB	13.32 kB	World routes only
Landing	21.64 kB	6.39 kB	Portal only
CSS	55.48 kB	10.37 kB	Always
ParticleField	2.47 kB	1.28 kB	Shared by both pages

Core Workflows

A · Entering a world

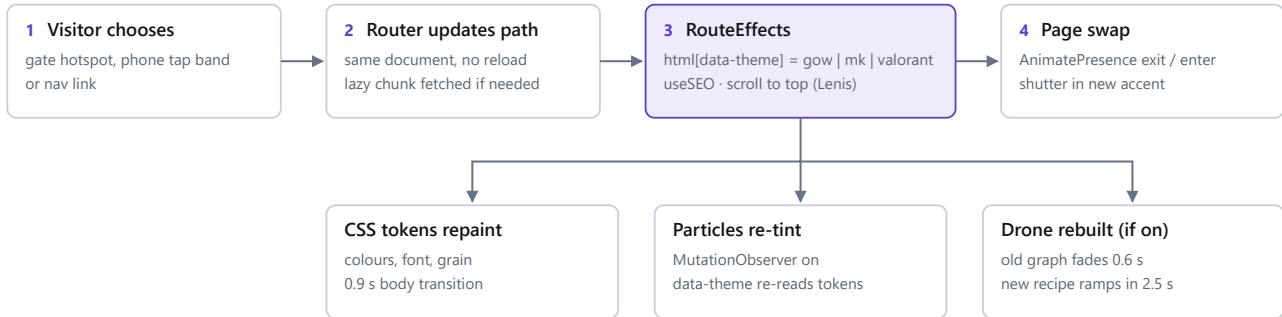
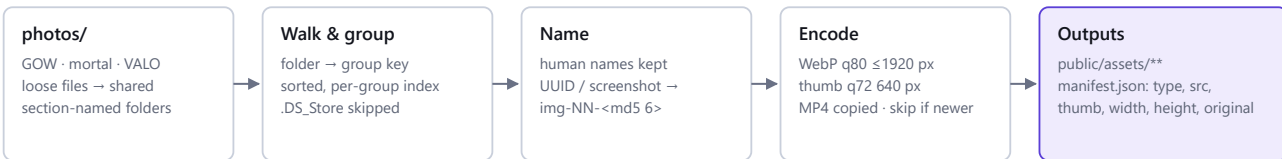


Figure 2 — One attribute write fans out to styling, canvas particles and audio. Verified: clicking a nav link changed the path and theme with no reload (a marker set on the window survived the navigation).

B · Media: from source file to screen



Runtime: assets.js → img(group, i) falls back to the group's last image (dev warning) · imgNamed(group, slug) returns undefined so callers can style around missing art
→ games.js composes content → LazyImage renders with intrinsic width/height from the manifest (verified CLS ≤ 0.0073 on every route)

Figure 3 — The asset pipeline. Re-run for this report: 118 of 118 entries were up to date and none were re-encoded (section 12).

C · When a video may make sound

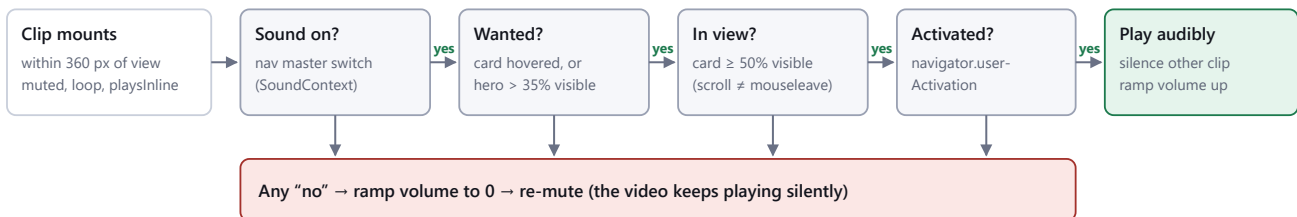


Figure 4 — Audio gating in [LazyVideo](#) and [Hero](#). The in-view condition was added on 11 Aug 2026 (commit 98c7aab) after a hovered card kept playing sound when scrolled off screen.

Important Engineering Decisions

D1 Theme through CSS variables selected by the route

PROBLEM	Four identities (portal plus three worlds) must change completely on navigation, without per-component conditionals.
DECISION	Scope nine tokens under <code>[data-theme]</code> and set that attribute from the pathname.
IMPLEMENTATION	<code>RouteEffects</code> writes <code>dataset.theme</code> ; styles and utilities read <code>var(--...)</code> ; canvas particles re-read tokens through a MutationObserver.
RESULT	Adding a palette is a CSS block. Verified: each route renders its own <code>--accent</code> and display font, and client-side navigation repaints without a reload.

D2 A section registry driven by per-world configuration

PROBLEM	The worlds present sections in different orders, and one section needed removing from a world without deleting its code.
DECISION	<code>GamePage</code> contains only a key → component map; order lives in <code>game.sections</code> .
IMPLEMENTATION	Unknown keys return <code>null</code> with a development warning; a default order covers configs without the field.
RESULT	Verified: all three rendered orders match the configuration. <i>Trade-off</i> : removing a key leaves the component dormant but still bundled (BossFights).

D3 Fault-tolerant, manifest-based asset lookup

PROBLEM	Deleting a source file rennumbers its group on the next pipeline run. According to code comments, a throwing lookup took the whole page down, and hard-coded map indices resolved two maps to the same image, producing a duplicate React key.
DECISION	Prefer a wrong image over a blank site, and look optional art up by name.
IMPLEMENTATION	<code>img()</code> falls back to the group's last image with a development warning; <code>imgNamed()</code> returns <code>undefined</code> ; map names pair with however much art exists via <code>slice()</code> .
RESULT	Verified: zero console errors and zero failed requests on all four routes of the production build.

D4 Commit optimised media; keep source media out of git

PROBLEM	339 MB of source media; the host should not run image processing at build time.
DECISION	Generate WebP at two sizes plus a manifest locally, commit the output, and git-ignore <code>photos/</code> .
IMPLEMENTATION	Incremental <code>npm run assets</code> based on file mtimes; <code>sharp</code> is a devDependency only.
RESULT	The production build needs no native image tooling. <i>Trade-offs</i> : a 125 MB <code>public/assets</code> in the repository, and no pruning of orphaned outputs (section 17).

D5 Drive Lenis from GSAP's ticker

PROBLEM	Smooth scroll and ScrollTrigger each want their own animation frame loop and fight over scroll position.
DECISION	Use one loop: GSAP's ticker calls <code>lenis.raf</code> , and Lenis scroll events call <code>ScrollTrigger.update</code> .
IMPLEMENTATION	<code>useSmoothScroll</code> with <code>lagSmoothing(0)</code> ; the instance is shared on <code>window.__lenis</code> for programmatic scrolling. The hook does nothing under reduced motion.
RESULT	Verified: Lenis is active at full motion and absent under <code>prefers-reduced-motion: reduce</code> .

D6 Scroll-spy via IntersectionObserver, built from the live DOM

PROBLEM	Lenis smoothing lags <code>window.scrollTo</code> , so a position-based spy flickered between neighbouring sections; stops for absent sections would be dead controls.
DECISION	Observe sections with a $\pm 45\%$ root margin, and build the rail from the sections present, sorted by document position.
RESULT	Verified on all three worlds: the rail lists only existing sections, in page order.

D7 Consent-first, single-source audio

PROBLEM	Browsers block unmuted autoplay; several clips could compete; sound on load is unwelcome.
DECISION	One master switch; a clip is audible only when every condition in Figure 4 holds; at most one audible clip at a time.
IMPLEMENTATION	<code>SoundContext</code> , a module-level <code>audibleVideo</code> singleton, volume ramps, <code>navigator.userActivation</code> checks; ambience synthesised with Web Audio.
RESULT	No audio assets to ship or license. Verified: the switch toggles state and label without errors. Audible output was not measured.

D8 Facades for third-party embeds

PROBLEM	A YouTube iframe on every world page loads third-party script whether or not anyone watches.
DECISION	Show a poster and play button, and create the iframe on click using the no-cookie host.
RESULT	Verified: zero iframes before the click, one after, served from <code>youtube-nocookie.com</code> .

D9 A dedicated phone composition instead of cropping the desktop layout

PROBLEM	The landscape portal plate side-cropped on phones, its painted gates were too small to tap, and header controls pushed the menu button off screen.
DECISION	Use a supplied portrait plate with full-width tap bands, move secondary controls into the menu sheet, and turn heroes into banners with copy beneath.
RESULT	Delivered across commits a37ee20 and 99e377f. Verified at 390 px: no horizontal overflow on any route, three tap bands, and the controls present in the sheet.

D10 Present supplied art-only plates without overlays

PROBLEM	Readability scrim washed out plates that already carry their own wordmark, and the visible title then existed only inside an image.
DECISION	An <code>artOnly</code> mode skips scrim and overlaid copy, adds a screen-reader <code>h1</code> , and reveals the single call to action during the clip's closing seconds.
RESULT	Verified: the Valorant hero exposes "Valorant — Defy The Limits" as an <code>h1</code> . <i>Side effect:</i> the Valorant signal-noise layer, also gated on <code>!artOnly</code> , never renders.

AI-Assisted Development

The repository shows that Claude Code was part of the development workflow. It does not record which lines were written by hand and which were generated, so this section reports only what the evidence supports.

Evidence

ARTEFACT	WHAT IT SHOWS
<code>.claude/launch.json</code> in the initial commit (ca92203, 7 Aug 2026)	Claude Code preview-server configurations for <code>npm run dev</code> (5173) and <code>npm run preview</code> (4173). Development ran inside Claude Code with a live preview of the app.
<code>scripts/contact-sheet.mjs</code>	Its header says it builds numbered grids "so a human (or Claude) can identify UUID-named images". <code>games.js</code> notes that character images were "identified from contact sheets". AI was used to identify visual assets and map them to configuration.
Git history (4 commits, 7–11 Aug 2026)	One initial commit of 325 files and 10,367 insertions, then three focused follow-ups on mobile navigation, mobile hero geometry, audio gating and iOS autoplay. All are authored by the project owner, and every message reads "Initial commit".
Code comments (581 comment lines in <code>src</code>)	Comments record the reason for decisions and the cause of fixed bugs (duplicate React key, washed-out scrims, scroll-spy flicker, iOS muted attribute). They document intent but do not identify who wrote them.
Development transcripts	Not available. Commits were made on a different machine, and this workspace is an extracted archive (a <code>_MACOSX</code> folder is present). The only earlier session on this machine was a brief project check on 7 Aug 2026.

How AI was used

Supported by evidence • Implementation inside Claude Code with a running preview server. • Identifying and naming unlabelled images from generated contact sheets. • Iterative fixes after the first build, such as audio continuing off screen and iOS autoplay. • Recording rationale for decisions and fixes in code comments.	Owner responsibilities • Concept and scope: a portal and three worlds. • Sourcing and organising all media, including section-level design plates. • Deciding what was committed; all four commits are the owner's. • Reviewing the running result, and responsibility for the shipped project.
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Validation of AI-assisted code. The follow-up commits are fixes found by using the app, not only by reading code, which shows why behaviour checks matter. The verification for this report (section 12) found two issues that were not visible in the source at a glance: a configuration flag read under the wrong name, and a hero effect that can never render.

How this report was produced

This report was prepared with Claude Code at the owner's request. Claude Code reviewed the code and history, ran the production build, wrote and ran the verification and measurement scripts, re-ran the asset pipeline and restored the manifest to its committed content, and captured the screenshots. No project source file was changed; `git status` shows no content differences. The owner is responsible for reviewing its claims, particularly sections 09 and 15, which describe their own role.

Implementation Details

Short excerpts from the mechanisms that carry the design, quoted from the repository.

src/index.css – one token set per world

```
[data-theme="gow"] {
  --bg: #05080d;  --bg-2: #0d1622;  --bg-3: #16273a;
  --accent: #c8442f;  --accent-2: #8fc7e8;
  --text: #eef2f6;  --muted: #8fa3b5;
  --font-display: "Cinzel", serif;
  --grain: 0.05;
}
```

src/App.jsx – the route selects the theme

```
function themeForPath(pathname) {
  const slug = pathname.replace(/^\//, '')
  return gameBySlug(slug)?.theme ?? 'home'
}
useEffect(() => {
  document.documentElement.dataset.theme =
    themeForPath(pathname)
  window.__lenis?.scrollTo(0, { immediate: true })
  window.scrollTo(0, 0)
}, [pathname])
```

src/hooks/useSmoothScroll.js – one frame loop

```
const lenis = new Lenis({ duration: 1.15, smoothWheel:
true })
lenis.on('scroll', ScrollTrigger.update)
const tick = (time) => lenis.raf(time * 1000)
gsap.ticker.add(tick)
gsap.ticker.lagSmoothing(0)
window.__lenis = lenis
```

src/sections/Hero.jsx – iOS muted-autoplay fix (99e377f)

```
<video ref={el} => {
  videoRef.current = el
  // React sets `muted` only as a property; iOS Safari
  // reads the attribute. defaultMuted reflects onto
  it.
  if (el) el.defaultMuted = true
}} autoPlay muted loop playsInline />
```

src/pages/GamePage.jsx – registry, no layout

```
const order = game.sections ?? DEFAULT_ORDER
return order.map((key) => {
  const Section = SECTIONS[key]
  if (!Section) {
    if (import.meta.env.DEV)
      console.warn(`GamePage: unknown section
"${key}"`)
    return null
  }
  return <Section key={key} game={game} />
})
```

src/data/assets.js – a wrong picture beats a blank site

```
export function img(key, i = 0) {
  const list = images(key)
  const item = list[i]
  if (item) return item
  if (import.meta.env.DEV) console.warn(`/* ... */`)
  return list[list.length - 1]
}
```

src/components/ui/LazyVideo.jsx – audio gate (98c7aab)

```
// Scrolling doesn't fire mouseleave, so gate on view.
const io = new IntersectionObserver([e] =>
  setInView(e.isIntersecting && e.intersectionRatio >=
0.5),
  { threshold: [0, 0.5, 1] })

if (audioActive && soundOn && inView) {
  if (navigator.userActivation &&
    !navigator.userActivation.hasBeenActive) return
  silenceOthers(v); audibleVideo = v
  v.muted = false; v.volume = 0
  rampVolume(v, volume)
} else if (!v.muted) rampVolume(v, 0)
```

Procedural ambience

`useAmbientAudio` builds a drone from a sine root, a triangle fifth detuned by 6 cents, and band-passed brown noise swept by a 0.05 Hz LFO. Recipes per theme:

THEME	ROOT	FIFTH	NOISE BAND	Q	GAIN
home	110 Hz	164.81 Hz	320 Hz	0.9	0.050
gow	73.42 Hz	110 Hz	180 Hz	1.6	0.060
mk	61.74 Hz	92.5 Hz	140 Hz	2.4	0.055
valorant	98 Hz	146.83 Hz	900 Hz	0.6	0.045

Other implementation notes

- **Portal geometry.** `useHeroGeometry` measures the frame with a `ResizeObserver`, fits the plate below the navbar, and converts each gate's fractional centre and size into pixels, so hotspots stay aligned with the painted gates at any viewport.
- **Particle budget.** `ParticleField` scales particle count with viewport width (minimum 24) and caps device pixel ratio at 2; `PortalWeather` starts and stops its loop with an `IntersectionObserver`.
- **Surface language.** `.frost`, `.glass`, `.panel` and `.metal-text` are built from tokens with `color-mix()`, so each world inherits them in its own colours.
- **Code splitting.** `manualChunks` puts framer-motion, gsap and lenis in one chunk and the React libraries in another.

Challenges & Solutions

Problems recorded in the commit history or in code comments, with the fix applied. The source of each is given.

CHALLENGE	SOLUTION	SOURCE · STATUS
Header controls pushed the menu button off a phone screen	Settings and Fullscreen hidden below 640 px and surfaced as buttons in the mobile sheet	Commit a37ee20 (10 Aug) · VERIFIED
Portal plate side-cropped on phones, hiding the wordmark and gates	First, width-fit geometry on narrow screens; then a dedicated portrait plate with full-width tap bands	a37ee20 → 99e377f (11 Aug) · VERIFIED
A hovered clip kept playing audio after being scrolled off screen	Second IntersectionObserver; audio requires ≥50% visibility	Commit 98c7aab (11 Aug) · NOT AUDIO-TESTED
Hero clips did not autoplay on iOS Safari	Set <code>defaultMuted</code> so the <code>muted</code> attribute is present	Commit 99e377f · NO DEVICE TEST
Hero blurb crossed the subjects on desktop but was wanted on phones	<code>hideDescription</code> hides it only from the <code>sm</code> breakpoint up	Commit 99e377f · OBSERVED
Removed source files broke references and crashed the page	Tolerant <code>img()</code> , name-based <code>imgNamed()</code> , positional map pairing	Comments in <code>assets.js</code> , <code>games.js</code> · NO ERRORS
Masonry gallery left ragged gaps	Uniform 4:3 grid; the lightbox shows each image whole	Comment in <code>Gallery.jsx</code> · VERIFIED
Scroll-spy flickered under smooth scrolling	IntersectionObserver in place of scroll maths	Comment in <code>JourneyRail.jsx</code> · VERIFIED
Readability scrimms washed out supplied artwork	<code>artOnly</code> mode with no scrimms and a screen-reader heading	Comment in <code>Hero.jsx</code> · VERIFIED
Deep hero zoom pushed subjects past the frame edges	Zoom limited to 1.04x; portal drift is scale-only	Comments in <code>Hero.jsx</code> , <code>index.css</code> · OBSERVED
The only download link could be hidden if the hero clip failed	A video <code>error</code> event reveals the call to action	Comment in <code>Hero.jsx</code> · NOT FAULT-INJECTED

Encountered while preparing this report

ISSUE	CAUSE	HANDLING
Dev server would not start on Windows	The extracted workspace copy of <code>node_modules</code> held macOS binaries; it is git-ignored, not committed	Reinstalled with <code>npm install</code>
Pipeline re-run rewrote all 118 manifest entries	The <code>original</code> field uses <code>path.relative</code> , which gives backslashes on Windows	Recorded as a limitation; manifest restored to its committed content
Default port 5173 already in use	Another local project	Used <code>--port</code> overrides (5199 dev, 4199 preview)

Testing & Verification

The repository contains no automated tests — no unit, component or end-to-end suite, no ESLint or Prettier configuration, and no CI. The results below come from verification carried out for this report and are not part of the project's tooling.

Method

- **Target:** production build (`vite build`) served by `vite preview` on localhost.
- **Driver:** headless Chrome controlled over the DevTools Protocol by a Node script; cache disabled; YouTube requests blocked so the run stayed local.
- **Viewports:** 1440 × 900 desktop, and 390 × 844 with touch emulation.
- **Evidence collected:** DOM state, computed CSS, console output, network events, layout-shift entries and screenshots.

Results — 69 of 71 checks passed

AREA	CHECKS	PASS	WHAT WAS EXERCISED
Route health	8	8	No console errors or exceptions; no failed or 4xx/5xx requests on all four routes
Layout	8	8	No horizontal overflow at 1440 px; CLS below 0.1 on every route
Theme	4	4	Correct <code>data-theme</code> , accent and display font per route
Configuration	9	7	Section order ×3, no boss section ×3, Valorant map captions hidden · fails: agent captions, signal-noise layer
SEO	7	7	Per-route title, canonical and <code>og:image</code>
UX & interaction	10	10	Portal hotspots; journey rail contents; lightbox open, next, wrap-around and close; sound switch
Performance behaviour	7	7	No iframe before click ×3; gallery clips not mounted before scrolling ×3; embed created on click
Security	4	4	<code>rel="noopener"</code> on external links ×3; no-cookie YouTube host
Accessibility & motion	5	5	Screen-reader <code>h1</code> ; Motion switch; Lenis and cursor off under reduced motion, on at full motion
Routing	2	2	Same-document navigation repaints theme; unknown route renders portal
Mobile	7	7	No overflow ×4; portrait plate with 3 tap bands; no custom cursor on touch; menu sheet controls
Total	71	69	

The two failures

CHECK	OBSERVED	CAUSE
Valorant agent cards hide DOM captions	15 of 15 cards render role and name	<code>Characters.jsx:158</code> passes <code>game.hideAgentLabels</code> ; config defines <code>hideArtLabels</code> (<code>games.js:309</code>)
Valorant signal-noise layer renders	Absent	<code>Hero.jsx:412</code> requires <code>!artOnly</code> ; the Valorant hero is <code>artOnly</code>

Other verification

ACTIVITY	RESULT
Production build	Succeeded · 489 modules · 10.66 s (Vite-reported) · no warnings printed
Asset pipeline re-run	118 of 118 entries up to date, 0 re-encoded. Regenerated manifest identical except the <code>original</code> path separator on Windows.
Source integrity after testing	Manifest blob hash equals HEAD (<code>9f6c219</code>); no content differences in the working tree
Secrets scan	No credentials in tracked files; the only environment variable is the public <code>VITE_SITE_URL</code>

Not verified

Real iOS and Android devices (including the iOS autoplay fix), Safari and Firefox, audible audio output, the Fullscreen API, throttled-network behaviour, Lighthouse scores, and the live Vercel deployment.

Security, Reliability & Data Integrity

Security

CONCERN	POSITION
Attack surface	Static files only: no backend, API, authentication, forms or database. The source contains no <code>localStorage</code> , <code>sessionStorage</code> , cookies, <code>fetch</code> or XHR.
Secrets	None in tracked files. <code>.env*</code> is git-ignored; <code>VITE_SITE_URL</code> is a public origin.
Third-party origins	The code references Google Fonts (loaded on page load), <code>youtube-nocookie.com</code> (only after a click, with <code>strict-origin-when-cross-origin</code> referrer policy), and storefront URLs as plain links.
Outbound links	<code>target="_blank"</code> with <code>rel="noopener noreferrer"</code> and descriptive <code>aria-label</code> (verified).
Security headers	GAP <code>vercel.json</code> sets only caching. No Content-Security-Policy, X-Frame-Options, Referrer-Policy or Permissions-Policy.

Reliability

- **Missing media degrades rather than breaks:** index fallback, name lookups returning `undefined`, a composed portal hero if the key-art plate is missing, poster frames beneath every clip.
- **Nothing waits forever:** the loading screen finishes on `load` or after 3.2 s; the hero call to action appears if its clip fails.
- **Configuration errors are contained:** unknown section keys render nothing; unknown routes render the portal (verified); an unknown slug on `GamePage` redirects home.
- **No timers or loops left behind:** effects clear volume ramps and cancel animation frames, and disconnect observers on unmount.
- **Observed stability:** zero console errors and zero failed requests across all routes and interactions tested.

Data integrity (media and content)

MECHANISM	EFFECT	STATUS
Generated manifest	Paths, dimensions and source file for every asset come from the files themselves, not typed by hand	IN PLACE
Content-hash names	Noise-named files get stable output names across re-runs	IN PLACE
Intrinsic dimensions	Images reserve space before decoding (CLS 0.0015–0.0073 measured)	VERIFIED
Immutable caching	One-year cache on <code>/assets/*</code> . It depends on changed files getting new names; a file replaced under the same name would stay cached.	CAVEAT
Output pruning	Not implemented: 21 images and 1 video on disk are not in the manifest	GAP
Cross-platform manifest	The <code>original</code> field uses OS path separators, so regeneration on Windows changes 118 lines	GAP

Content and rights

The site presents three commercial franchises, and the supplied portal art includes their wordmarks. The project labels itself a non-affiliated fan concept, uses original glyphs for platform marks, and links to official storefronts. Any public deployment should respect the rights holders' fan-content policies.

Deployment & Usage

Local

```
npm install      # per platform; node_modules is
git-ignored
npm run dev      # Vite, http://localhost:5173
npm run build    # → dist/
npm run preview  # http://localhost:4173
npm run assets   # only after changing photos/
```

A fresh clone does not need `photos/`, because the optimised output and manifest are committed. If a default port is taken, pass `-- --port <n>`.

Vercel

```
{
  "rewrites": [{ "source": "/(.*)", "destination":
"/index.html" }],
  "headers": [{ "source": "/assets/(.*)",
  "headers": [{ "key": "Cache-Control",
    "value": "public, max-age=31536000, immutable"
  } ] } ]
}
```

Framework preset: Vite. Set `VITE_SITE_URL` to the real origin so canonical and Open Graph URLs are correct; it defaults to `https://gameverse.vercel.app`. A live deployment was not checked for this report.

Adding a fourth world

- 1. Place media in `photos/<NAME>/` using the existing sub-folders, add the folder to `GAME_DIRS` in the pipeline, and run `npm run assets`.
- 2. Add a `[data-theme="..."]` token block to `index.css`, and a drone recipe to `useAmbientAudio`.
- 3. Add an entry, including `sections`, to `games.js`.
- 4. Add the nav link and a sitemap URL.

No page or section component needs to change.

Repository contents

TRACKED

`src/` (45 files incl. manifest), `scripts/`, `public/` (270 files), configs, README, `.claude/launch.json`, and a stray root `Baldur.png` (2.4 MB, unreferenced)

IGNORED

`node_modules/`, `dist/`, `photos/`, `.env*`, logs, editor folders, `.DS_Store`

My Role & Contribution

Gopi Patel owns the project and authored every commit. The table separates what the repository shows directly from what it cannot show.

AREA	CONTRIBUTION	EVIDENCE
Concept & scope	Defined the product: one portal, three worlds, and a section set for each	README; top-level media folders per world
Art direction & design references	Supplied section-level design plates, including a whole-page mockup, and a dedicated phone plate for the portal	Source names (<i>hero section, brutality section, game mode section, home page for mobile</i>); comment "the supplied plate ... is a whole page mockup"
Media sourcing & curation	Collected and organised 339 MB of source media, from which 106 images and 12 clips are used	<code>photos/</code> structure; manifest <code>original</code> fields; filenames dated 20 Jul – 5 Aug 2026
AI-assisted implementation	Developed the application in Claude Code with a live preview, and iterated on mobile layout, audio behaviour and iOS playback	<code>.claude/launch.json</code> ; commits a37ee20, 98c7aab, 99e377f
Quality & acceptance	Reviewed the running app and decided what shipped	Follow-up commits fixing issues found by using the app
Delivery setup	Version control and hosting configuration	Git history; <code>vercel.json</code> ; <code>.env.example</code> ; README

What the evidence does not show. Line-level authorship. Claude Code was part of the implementation workflow, and the repository does not separate hand-written code from generated code, so this report makes no claim of manual authorship for specific modules. The owner's contribution is shown through direction, media and design inputs, iteration, and acceptance.

Skills demonstrated

Product & design

Defining a multi-world experience; preparing section-level design references; separate compositions for desktop and phone.

Directing AI-assisted engineering

Turning a visual brief into a working React application with Claude Code; finding and fixing behaviour issues across mobile, audio and iOS.

Delivery

A static deployment design with committed optimised media, per-route SEO and cache configuration.

Results & Verified Metrics

69 / 71

VERIFICATION CHECKS

0

CONSOLE ERRORS, 4
ROUTES

≤ 0.0073

CLS ON LOAD

0 px

OVERFLOW AT 390 & 1440

First-load transfer per route

ROUTE	REQUESTS	TOTAL	SCRIPT	CSS	IMAGES	FONTS	VIDEO*
/	17	847 kB	193 kB	12 kB	568 kB	73 kB	—
/god-of-war	29	8,902 kB	199 kB	11 kB	989 kB	97 kB	7,605 kB
/mortal-kombat	23	2,152 kB	199 kB	11 kB	1,853 kB	87 kB	—
/valorant	25	2,946 kB	199 kB	11 kB	811 kB	72 kB	1,851 kB

Measured against [vite preview](#) on localhost, cache disabled, Chrome at 1440 × 900, sampled 7 s after navigation with no scrolling. *Video was still streaming, so these figures are partial. They show relative weight, not field performance.

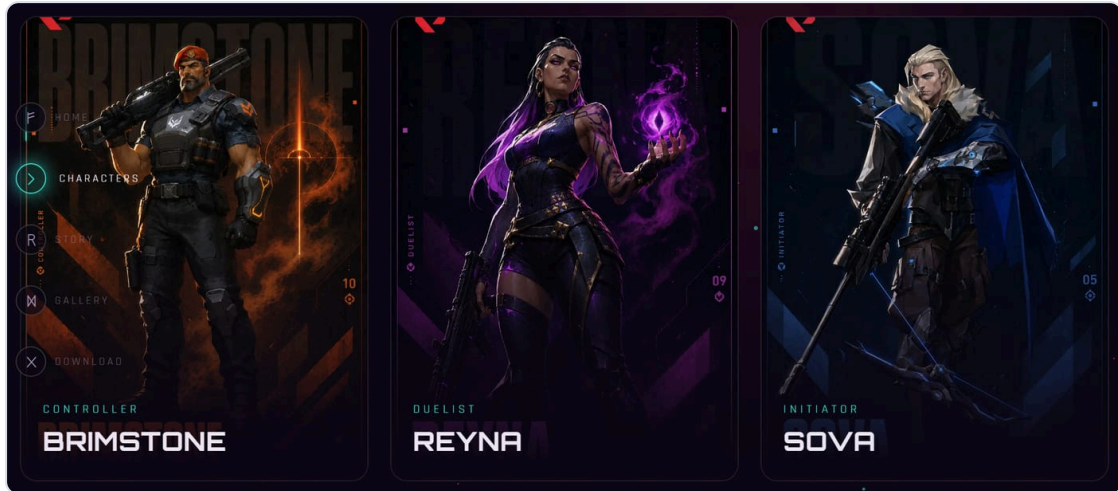
Video mounted on first load (no scrolling)

ROUTE	CLIPS MOUNTED WITH PRELOAD="AUTO" AND AUTOPLAY	FILE SIZE
/god-of-war	Hero clip + all 5 character clips; the character rail sits inside the 360 px mount margin below the hero	30.2 + 12.1 MB
/valorant	Hero clip	29.1 MB
/mortal-kombat	None; no local clips	—
All routes	Gallery clips: 0 mounted before scrolling (verified)	—

Outcomes against objectives

OBJECTIVE	OUTCOME	ASSESSMENT
One entry point, distinct worlds	Theme, font, particles and section order change per route with no reload	MET
Follow the supplied art direction	Plates shown at their own aspect ratios; art-only mode; captions suppressed on maps	MOSTLY
Add worlds without page code	Registry plus configuration; verified orders	MET
Media usable on phones and ordinary links	Lazy mounting, facades and WebP thumbnails work; 30 MB hero clips and eager character clips remain	PARTLY
No forced motion or sound	Reduced motion honoured structurally; sound only after consent	MET
Static deployment with no build-time image work	Build succeeds without image tooling; hosting configured	MET

Limitations



Verified defect. Each supplied agent card already carries the agent's name and role. Because `Characters` reads the wrong flag, the DOM caption is printed on top of the baked-in name (visible behind "BRIMSTONE").

#	LIMITATION	DETAIL	SEVERITY
1	Valorant caption flag mismatch	<code>hideAgentLabels</code> read, <code>hideArtLabels</code> set; 15 cards affected	VISIBLE
2	Video weight	Two hero clips of about 30 MB; on <code>/god-of-war</code> , all character clips also start downloading immediately. No phone-specific, low-bitrate or Save-Data path.	HIGH
3	No automated tests, lint or CI	Regressions such as #1 and the duplicate key in #5 are not caught automatically	HIGH
4	Dormant or unused code	BossFights (218 lines) and 3 boss entries; Valorant signal-noise layer; StatsBand (133, with placeholder figures) and TrailerStrip (173); a 581-line fallback hero used only if the plate is missing	MEDIUM
5	Configuration hygiene	The God of War entry declares <code>sections</code> twice (identical values, so the second silently wins); three of four Valorant map names are unused	LOW
6	Client-side-only SEO	Crawlers that don't run JavaScript see identical base tags for all routes; no prerendering	MEDIUM
7	Asset pipeline gaps	No pruning (21 images + 1 video orphaned); OS-specific <code>original</code> paths	LOW
8	Lightbox accessibility	No <code>role="dialog"</code> / <code>aria-modal</code> , no focus trap or focus return, and the enlarged image has <code>alt=""</code>	MEDIUM
9	No security headers	No CSP or related headers in <code>vercel.json</code>	LOW
10	Documentation drift	README describes a masonry gallery, <code>preload="none"</code> clips and a ~230 MB source folder; code uses a grid and <code>metadata</code> / <code>auto</code> , and the folder measures 339 MB. Commit messages don't describe changes.	LOW
11	Verification coverage	No real-device, Safari/Firefox, audio-output or live-deployment testing (section 12)	MEDIUM
12	Repository weight	125 MB of committed media, plus a stray 2.4 MB <code>Baldur.png</code> at the root	LOW
13	Rights	Franchise names and wordmarks in supplied art; fan-concept status must be respected in any public use	CONTEXT

Future Improvements

Ordered by the ratio of impact to effort, based on the verified state of the project.

P	IMPROVEMENT	CONCRETE CHANGE	EFFORT
1	Fix the caption defect	Read <code>game.hideArtLabels</code> in <code>Characters.jsx</code> ; delete the duplicate <code>sections</code> key	Minutes
2	Make verification permanent	Port the 71 checks to Playwright; add ESLint (<code>no-dupe-keys</code> , unused imports); run both in CI on every push	1–2 days
3	Reduce video cost	Re-encode hero clips at lower bitrates; show the poster only on narrow screens or with Save-Data; use <code>preload="metadata"</code> for character clips and mount them on actual visibility	1–2 days
4	Resolve dormant code	Either add <code>bosses</code> to the God of War order and fix the signal-noise gate, or delete them with StatsBand and TrailerStrip	Hours
5	Harden the pipeline	<code>--clean</code> pruning of orphaned outputs; POSIX separators in <code>original</code> ; a check that every manifest entry exists on disk	Hours
6	Accessible lightbox	Dialog role, <code>aria-modal</code> , focus trap and return, meaningful alt text	Hours
7	Prerender routes	Emit static HTML for the four routes so link previews and non-JS crawlers see per-route metadata	Half a day
8	Security headers	CSP allowing self, Google Fonts and the no-cookie YouTube frame; Referrer-Policy; Permissions-Policy	Hours
9	Real-device testing	iOS Safari (autoplay fix), Android Chrome, desktop Safari and Firefox; audio behaviour by ear	Half a day
10	Documentation	Update the README to match the code; use descriptive commit messages going forward	Hours

Conclusion

GAMEVERSE delivers what it set out to do: one portal, three worlds, and a visual identity that changes completely on navigation while the underlying pages stay generic.

Its strongest parts are architectural. A single attribute drives four themes. Configuration decides what each world shows and in what order. Media is addressed through a generated manifest that tolerates missing files. On the production build every route loaded without console errors or failed requests, kept layout shift below 0.01, fitted phone and desktop widths without overflow, and honoured reduced motion.

Verification also found real work remaining. A mistyped flag duplicates captions on the Valorant roster. Two effects never render. Hero video is heavy for phones. There is no automated testing to catch these regressions. None of these issues is architectural, and the first can be fixed in minutes.

The project was built in an AI-assisted workflow with Claude Code, directed by the owner, who supplied the concept, media and design references and accepted each change. The commit history shows iteration driven by using the app: mobile layout, audio behaviour, iOS playback. This report's verification continues that approach.

Master Prompt

A single prompt that rebuilds this project from an empty folder. It follows the tagged structure — role, goal, context, instructions, constraints, input, examples, output format, final check — which also matches the universal order, so it transfers to other models by renaming the tags to their headings. Supply your own artwork and clips in `photos/`; no prompt can recreate media.

<role>

You are a senior front-end engineer and creative developer. You build production-quality React applications from a written brief, and you are fluent in scroll-driven animation, canvas rendering, the Web Audio API, responsive art direction and image build pipelines.

</role>

<goal>

Build GAMEVERSE: a static, client-side React single-page application. A cinematic portal page opens into three long-scroll "world" pages – God of War, Mortal Kombat and Valorant. Moving between them changes the entire identity (palette, display typeface, film grain, particle weather, ambient audio, page-transition colour) while every page and section component stays generic and content-driven.

Done means: the repository builds with no warnings, an automated test suite passes against the production build, and the site deploys to static hosting with no server and no build-time image processing.

</goal>

<context>

- Audience: visitors browsing three game worlds on desktop and phone. There is no backend, no account system, no form and no user data.
- This is a non-affiliated fan concept. Say so in the footer and the README. Draw platform and social marks as original geometric SVG paths; never reproduce a company's logo.
- I supply all artwork and video. Nothing is generated or downloaded by the build.
- Hosting: Vercel, static output, SPA rewrite. The host must never run image tooling.
- I will review the running site, not only the code, so the app must be correct in a browser: no console errors, no layout shift, no sideways scrolling, and no surprise audio.

The media I provide lives in `photos/` (git-ignored) in exactly this shape:

`photos/`

<code>hero-fullscreen.png</code>	landscape portal key art: three lit gates, wordmark baked in
<code>home page for mobile.png</code>	portrait portal key art: the same three gates stacked
<code>gow-hero.png, *.png</code>	loose plates

`GOW/`

<code>background for hero section/</code>	<code>gow-hero-poster.jpg + upscaled-video.mp4</code> (hero clip)
<code>characters/</code>	kratos, atreus, freya, baldur, odin, tyr, heimdall
<code>characters/videos_/</code>	5 short character clips
<code>gallery/ , gallery/videos/</code>	15 stills, 4 clips
<code>story_nd_features.png</code>	

`mortal/`

<code>hero section.png, brutality section.png, game mode section.png</code>	
<code>characters/</code>	14 fighter portraits
<code>gallery_/</code>	18 stills

`VALO/`

<code>valo-hero-poster.jpg, upscaled-video.mp4, hero-video.mp4</code>	
<code>characters_/</code>	15 agent cards that ALREADY show the agent name and role
<code>gallery/</code>	14 stills
<code>maps/new.png</code>	one wide plate showing the whole map pool

Some filenames are UUIDs or screenshots. Do not guess what they depict: build the contact-sheet

helper described below, show me numbered grids, and ask me to name them.

</context>

<instructions>

1. Scaffold the project.

React 19 + Vite 6 in JavaScript (no TypeScript). react-router-dom 7, framer-motion 12, gsap 3 with ScrollTrigger, lenis 1.3. Tailwind CSS v4 through @tailwindcss/vite, configured only by a single @import "tailwindcss" in src/index.css – no tailwind.config file. sharp as a devDependency only. Add ESLint (flat config, react + react-hooks, no-dupe-keys) and Playwright before writing feature code. Scripts: dev, build, preview, assets, lint, test.

2. Build the theme system first, because everything else depends on it.

Define the token set under [data-theme="home" | "gow" | "mk" | "valorant"] in src/index.css: --bg, --bg-2, --bg-3, --accent, --accent-2, --text, --muted, --font-display, --grain (values in <input>). Body font Rajdhani; load the five families from Google Fonts with preconnect and display=swap. Give body a 0.9s cubic-bezier(.16,1,.3,1) colour transition.

Build the shared surface language from tokens with color-mix(), so each world inherits it in its own colours: .frost (frosted glass, 16px blur, an --accent-2 hairline along the top edge via ::before, lift and brighter rim on hover), .glass, .panel, .metal-text (a five-stop gradient clipped to the glyphs, bloom via drop-shadow, not text-shadow), .hairline, .glow-ring, .text-glow, and a fixed .grain::after overlay using an inline SVG feTurbulence filter at opacity var(--grain).

3. Make the route select the theme.

App.jsx derives a theme slug from the pathname and writes it to document.documentElement.dataset.theme on every navigation, then resets scroll (through Lenis when it is running, and window.scrollTo as a fallback). One attribute write must repaint the whole site.

4. Put all content in configuration.

src/data/games.js exports one array; each world has: slug, theme, name, titleLines[2], tagline, effect (snow|ember|neon), sections[] (render order), hideArtLabels, hero{bg, video?, description, align?, artOnly?, hideDescription?, hideTrailer?}, story{kicker, title, bg, aspect?, paragraphs[]}, features[5]{icon, title, text}, characters[]{name, role, image, video?}, maps[]?, gallery[], trailer{youtube, poster, video?, note}, requirements{minimum[], recommended[]}, download{headline, sub, platforms[]{name, url}}.

src/pages/GamePage.jsx holds only a SECTIONS registry (key to component) and maps game.sections over it. An unknown key logs a warning in development and renders null; an unknown slug redirects to "/". GamePage contains no layout of its own.

5. Write the asset pipeline (scripts/optimize-assets.mjs, npm run assets).

Walk photos/; map GOW to gow, mortal to mk, VALO to valorant, loose files to shared; derive a group key per output folder (lowercase, spaces to dashes); keep human filenames, and turn UUID or "Screenshot" names into img-NN-<6-character content hash> so names survive re-runs; write WebP at max 1920px wide q80 plus a 640px q72 .thumb; copy MP4s unchanged; skip any output newer than its source; support --clean to delete outputs no longer in the manifest.

Emit src/data/manifest.json entries: {type, src, thumb, width, height, original}, always with forward slashes so the file is identical on Windows, macOS and Linux.

Also write scripts/contact-sheet.mjs <group> <out.png>: a numbered five-column thumbnail grid used to identify unlabelled art.

Commit public/assets and the manifest. Git-ignore photos/, node_modules, dist and .env*.

6. Wrap media access in one module (src/data/assets.js) that is the only place aware of paths:

group, images, videos, img(key, i), imgNamed(key, slug), vid, vidNamed. img() with an out-of-range index returns the group's last image and warns in development – it must never throw, because a wrong picture is recoverable and a blank page is not. imgNamed() returns undefined so callers can style around missing optional art. Pair map names with however much map art exists using slice(), so two names can never resolve to the same image.

7. Build the persistent shell.

BrowserRouter; React.lazy for Landing and GamePage behind a Suspense spinner; AnimatePresence mode="wait" with a PageTransition that fades and sweeps a 105-degree shutter in the destination theme's accent. useSEO sets title ("Game" – GAMEVERSE), and

"GAMEVERSE – Enter Legendary Game Universes" on the portal), description, canonical, og:* and twitter:*, reading the origin from VITE_SITE_URL with a documented fallback.

LoadingScreen: wordmark plus a bar that ramps to 92% and completes on window load, with a 3.2s failsafe. Nav: Home, the three worlds, and an inert "Next Game · SOON" teaser with no route; the bar retracts when scrolling down past 140px but never while a menu is open; a glow follows the cursor along it; a sound button; a Settings popover with three real switches – Ambience, Motion (sets `html[data-motion="reduced"]`) and Fullscreen (mirrors `document.fullscreenElement` so Esc and F11 keep the label honest). Below 640px, Settings and Fullscreen leave the header and Motion + Fullscreen appear in the mobile sheet, which locks body scroll while open.

Cursor (instant dot, ring easing behind it, 2.1x over interactive elements) and MouseGlow (680px accent radial) mount only for (pointer: fine) and never under reduced motion.

ScrollProgress bar. JourneyRail at 1280px and wider: rune stops built from the sections actually present, sorted by document position, with an IntersectionObserver scroll spy (`rootMargin -45% top and bottom`) – never scroll arithmetic, which flickers under smooth scroll.

Footer: links, year, fan-concept disclaimer, back to top.

8. Build the portal page (/).

Desktop: cover-fit the landscape plate into the frame with a ResizeObserver geometry hook that seats the plate's top edge at the navbar's height, so the baked-in wordmark is never hidden. Convert each gate's fractional centre and size (see `<input>`) into pixels, and give every gate a breathing radial bloom, counter-rotating dashed SVG rings, its own small weather canvas and a Link labelled "Enter {name}". Add storm flashes on two different clocks, two fog bands drifting opposite ways, spring cursor parallax applied to one shared transform chain (so overlays can never slide out of register with the painted gates), a focus-pull dim on hover, and a 34s scale-only drift – never a pan, which would swing an edge out of frame.

Phones: render the portrait plate at full width, padded down by the navbar height, with three full-width tap bands over the stacked gates. If the desktop plate is missing from the manifest, fall back to a composed three-gate hero rather than failing.

9. Build the sections, each generic over a `game` prop.

Hero – video (with the still as its poster) or still; set `el.defaultMuted = true` on the video ref, because React sets muted only as a property while iOS Safari reads the attribute when deciding whether a muted autoplay may start. GSAP mouse parallax on `[data-depth]` layers; a 26s yoyo zoom to 1.04 (no deeper: the plates are already cropped to fill). Per-world atmosphere scoped so it cannot leak: for gow, drifting fog, a double-strike lightning flash, torchlight flicker and SVG ravens; for a cyber world, a scanline sweep, a chromatic tear that fires about twice a minute and an energy wash. When `hero.artOnly` is set, skip the readability scrim and the overlaid copy entirely (they wash out art that carries its own wordmark), add an sr-only h1 of "{name} – {tagline}", and show one Download call to action during the clip's closing three seconds – revealing it immediately if the video errors, so the only link is never stranded. On phones the hero art becomes a full-width banner at its own aspect ratio with the copy stacked beneath; `hideDescription` hides the blurb only from the sm breakpoint up.

Characters – horizontal snap rail with arrow buttons; cards tilt up to 5 degrees toward the pointer; clips autoplay muted once mounted and become audible on hover under the audio rules below; captions are suppressed when `hideArtLabels` is set, because that artwork is already labelled.

Story – scroll-linked parallax (y -8% to 8%, scale 1.15 to 1 to 1.15) with a drifting light band; a story may pin the panel to the artwork's own aspect ratio, and when it does, soften the parallax so the matched framing is not zoomed back into a crop.

Features – five cards, GSAP ScrollTrigger stagger (y 60, 0.8s, power3.out, 0.12 stagger, start "top 80%"), pointer tilt up to 3 degrees, inline SVG icons.

Maps – panels sized to each plate's own aspect ratio, with a holographic grid and a scan sweep on hover; a single plate takes the full width.

Trailer – a band that scales 0.9 to 1 on entry showing a poster and play button; create the `youtube-nocookie.com` iframe only on click, with `referrerPolicy strict-origin-when-cross-origin`. Loading an embed eagerly would pull roughly a megabyte of third-party script into every page.

Gallery – a uniform grid (2 / 3 / 4 columns) of 4:3 tiles using thumbnails; video tiles preview on hover; clicking opens a lightbox with `role="dialog"` and `aria-modal`, a focus trap, focus returned to the trigger on close, Escape to close, wrapping arrow keys, and real alt text.

SystemRequirements – minimum and recommended cards using `dl/dt/dd`, the recommended one ringed and badged.

Download – accent wash, floating platform glyphs, and storefront buttons that open in a new tab with `rel="noopener noreferrer"` and descriptive aria-labels.

10. Media, audio and motion behaviour.

LazyImage takes a manifest entry and renders with loading="lazy", intrinsic width and height from the manifest (so nothing shifts), and an accent wash held until decode.

LazyVideo mounts its element only within 360px of the viewport and prefers preload="metadata". A clip may be audible only when ALL of these hold: the nav sound switch is on; the parent reports hover (or, for a hero, it is more than 35% visible); the clip is at least 50% visible (scrolling never fires mouseleave, so hover alone would leave sound playing off screen); and navigator.userActivation.hasBeenActive is true. Otherwise ramp the volume to zero and re-mute. Ease volume rather than snapping it, and enforce with a module-level singleton that at most one clip is ever audible. SoundContext is the single master switch: when it is off, nothing may make noise.

useAmbientAudio(theme) synthesises the drone with Web Audio – no audio files to ship or license: a sine root, a triangle fifth detuned 6 cents for slow beating, and a looping brown-noise buffer through a bandpass swept by a 0.05Hz LFO. Gain ramps in over 2.5s and fades over 0.6s on change, so switching worlds cross-fades instead of clicking. Stay silent until the visitor enables it.

ParticleField is one fixed full-page canvas whose variant selects physics: dust 60 particles (gravity -0.14), snow 110 (+0.35, lateral sway), ember 80 (-0.5, flicker), neon 70 (-0.18, flicker plus rare streaks). Scale the count with viewport width, cap devicePixelRatio at 2, and re-read colours from the tokens through a MutationObserver on data-theme.

PortalWeather is a boxed canvas so three climates can run side by side; its loop must start and stop with an IntersectionObserver.

useSmoothScroll boots Lenis (duration 1.15, exponential easing) and drives it from GSAP's ticker with lagSmoothing(0), wiring lenis "scroll" to ScrollTrigger.update, so the two share one animation frame loop instead of fighting over scroll position. Expose the instance on window.__lenis for programmatic scrolling. Do not start it at all under reduced motion. Register ScrollTrigger exactly once, in src/animations/gsap.js, and import gsap from there.

11. Honour reduced motion twice over: a CSS media query and an [data-motion="reduced"] rule set both collapsing animation and transition durations to 0.01ms, plus a live matchMedia hook so components opt out structurally – MouseGlow returns null, Cursor never attaches, Lenis never boots, the feature stagger is skipped, tilt is disabled.

12. Performance and delivery.

vite.config.js: target es2020; manualChunks with motion: [framer-motion, gsap, lenis] and vendor: [react, react-dom, react-router-dom], so the animation layer caches independently of application code. Keep hero clips small – re-encode to a few MB, and on narrow screens or when Save-Data is set, show the poster instead of downloading the video.

vercel.json: rewrite /(.* to /index.html; set Cache-Control public, max-age=31536000, immutable on /assets/(.*); add Content-Security-Policy (self, Google Fonts, the no-cookie YouTube frame), Referrer-Policy and Permissions-Policy.

public/: favicon.svg, robots.txt, sitemap.xml with all four routes. Document VITE_SITE_URL in .env.example. Base SEO tags in index.html for scrapers that do not run JavaScript.

13. Write the Playwright suite against the preview build, covering at least the checks listed in <final_check>. Keep ESLint clean. Write a README describing the theme mechanism, the asset pipeline, the performance tactics, accessibility, SEO and deployment – and keep it accurate. Use descriptive commit messages.

</instructions>

<constraints>

- No TypeScript, no state-management library, no UI component kit, no icon library, no CSS-in-JS, no audio files, no analytics.
- No component may branch on which world is active in order to choose a colour, font or grain. Styling comes from the tokens only.
- Components must never hard-code an /assets path; they ask assets.js by group and index or name.
- One flag name per behaviour. hideArtLabels must be read by every section that draws a caption over supplied art (characters AND maps). Do not introduce a second spelling.
- Never gate a per-world effect on a flag that world also sets for another purpose; keep the condition for hero atmosphere independent of artOnly.
- Nothing may play audibly before the visitor enables sound, and never more than one clip at once.
- Every decorative layer is aria-hidden; every icon-only control has an aria-label; focus-visible

is always styled; semantic elements throughout (main, section, article, nav, footer, dl).

- Keep the committed repository buildable without photos/, because only the optimised output and the manifest are committed.
- Do not invent content: use the copy and data given in <input>, and ask me when art is ambiguous.
- Comment why, not what – especially where a browser quirk or a real failure forced the approach.

</constraints>

<input>

THEME TOKENS (bg / bg-2 / bg-3 · accent · accent-2 · text · muted · display font · grain)

home #05060a #0b0e1a #141a2e · #7c5cff · #00e0ff · #e8ecf4 · #8b93a8 · Orbitron · .035

gow #05080d #0d1622 #16273a · #c8442f · #8fc7e8 · #eef2f6 · #8fa3b5 · Cinzel · .05

mk #080303 #1a0709 #2e0d0b · #ff5a1f · #f5b942 · #f7ece5 · #ad8378 · Cinzel Decorative · .045

valorant #0a0512 #150b26 #221041 · #ff2d78 · #38e0d0 · #f0eaf7 · #9d8fb5 · Orbitron · .04

PORTAL GATES (fractions of the landscape plate: centre x, centre y, width, height)

god-of-war .193 .551 .263 .541 glow #4da6ff weather snow

mortal-kombat .501 .566 .254 .522 glow #ff7a1f weather ember

valorant .807 .557 .234 .517 glow #ff2d78 weather digital

PORTRAIT PLATE BANDS (top, height as fractions): .045/.31 · .355/.315 · .67/.33

AMBIENT AUDIO RECIPES (root, fifth, noise band, Q, gain)

home 110 / 164.81 / 320Hz / 0.9 / .05 gow 73.42 / 110 / 180Hz / 1.6 / .06

mk 61.74 / 92.5 / 140Hz / 2.4 / .055 valorant 98 / 146.83 / 900Hz / 0.6 / .045

GOD OF WAR – theme gow, effect snow, tagline "A Journey Through Norse Mythology".

Order: hero, characters, story, features, trailer, gallery, requirements, download.

Hero: the upscaled clip with its own poster frame; hide the trailer button (the cinematic has its own section); blurb about Kratos living as a man outside the shadow of the gods.

Characters: Kratos "The Ghost of Sparta", Atreus "The Archer", Freya "The Vanir Witch", Baldur "The Invulnerable", Odin "The All-Father" – each with a hover clip.

Story: kicker "The Story", title "A Journey of Father and Son", two paragraphs, square plate pinned to 1:1. Features: Epic Combat, Stunning Realms, Skill System, Puzzles, Upgrade & Craft.

Trailer: YouTube dIQGI36BxDE, note "Father and Son".

Requirements: min i5-6600K / 8GB / GTX 1060 6GB / 70GB; rec i7-4770K / 16GB / RTX 2070 / 70GB SSD.

Storefronts: Steam, Epic Games, PlayStation – deliberately no Xbox, because the series does not ship there and a button that leads nowhere is worse than no button.

MORTAL KOMBAT – theme mk, effect ember, tagline "Fight Your Destiny". Same section order.

Hero: centre the copy between the two fighters, and hide the blurb and second button on desktop because they crossed into the artwork; keep the blurb on phones.

Characters (name, epithet): Jade "Emerald Enforcer", Reptile "Saurian Stalker", Sub-Zero "Grandmaster of Frost", Raiden "God of Thunder", Shao Kahn "Konqueror of Realms", Ermac "Legion of Souls", Mileena "Empress of Ambition", Sindel "Queen of Screams", Frost "Cryomancer Ascendant", Scorpion "Vengeful Spectre", Kitana "Princess of Edenia", Liu Kang "Champion of Fire", Smoke "Living Vapor", Shang Tsung "Soul Sorcerer".

Story: kicker "Brutalities", title "Finish Him", wide plate pinned to 16:10 so both fighters show whole. Features: Story Mode, Brutalities, Legendary Fighters, Online Battles, Customization. Trailer: YouTube UZ6eFEjFfJ0, note "Kinematic reveal".

Requirements: min i5-6600 / 8GB / GTX 980 / 100GB; rec i7-8700K / 16GB / RTX 2060 / 100GB SSD.

Storefronts: Steam, Epic Games, PlayStation, Xbox.

VALORANT – theme valorant, effect neon, tagline "Defy The Limits", hideArtLabels true.

Order: hero, characters, story, maps, features, trailer, gallery, requirements, download.

Hero: artOnly – the plate carries its own wordmark, tagline and lighting, over the looping clip.

Agents (name, role): Brimstone Controller, Reyna Duelist, Sova Initiator, Breach Initiator, Killjoy Sentinel, Yoru Duelist, Omen Controller, Viper Controller, Skye Initiator, Chamber Sentinel, Astra Controller, Cypher Sentinel, Raze Duelist, Jett Duelist, Sage Sentinel.

Story: kicker "The Protocol", title "First Light Changed Everything", plate pinned to 16:9.

Features: Agents, Weapons, Maps, Ranked, Esports. Maps: name the plates you are given; never list more names than there is art. Trailer: YouTube lWr6dhTcu-E, note "Cinematic launch".

Requirements: min Core 2 Duo E8400 / 4GB / Intel HD 4000 / 20GB; rec i3-4150 / 8GB / GTX 1050 Ti / 20GB SSD. Storefronts: Riot Client, PlayStation, Xbox – it ships through Riot's own launcher.

```

</input>

<examples>
<example>
Input: I add photos/NEWWORLD/ with characters/ and gallery/, then add a token block and a games.js
entry with sections: ['hero', 'characters', 'gallery', 'download'].
Desired output: running npm run assets and adding a nav link is enough. /newworld renders those
four sections in that order, in its own palette and font, with its own particle variant and drone.
No page or section component is edited.
</example>

<example>
Input: I delete one image from photos/GOW/characters/ and re-run the pipeline, so a games.js entry
now points past the end of that group.
Desired output: the page still renders. img() returns the group's last image and logs a
development warning naming the group and index. Nothing throws, and no route shows a blank screen.
</example>

<example>
Input: A visitor hovers a character card with sound on, then scrolls the card off screen without
moving the pointer.
Desired output: as the card passes below 50% visibility, its volume ramps to zero and the element
re-mutes. The clip keeps playing silently. No other clip becomes audible.
</example>
</examples>

<output_format>
Deliver a repository in this shape:

index.html · package.json · vite.config.js · vercel.json · .env.example · eslint.config.js
README.md · .gitignore (node_modules, dist, photos/, .env*)
public/          favicon.svg · robots.txt · sitemap.xml · assets/** (generated, committed)
scripts/         optimize-assets.mjs · contact-sheet.mjs
tests/           Playwright specs
src/
  main.jsx · App.jsx · index.css
  animations/gsap.js · context/SoundContext.jsx · utils/cn.js
  data/          games.js · assets.js · manifest.json (generated)
  hooks/         useSmoothScroll · useAmbientAudio · useReducedMotion · useSEO
  components/    Nav · Footer · Cursor · MouseGlow · ScrollProgress · LoadingScreen
                  PageTransition · JourneyRail · ParticleField · PortalWeather
                  ui/ GlowButton · Icon · LazyImage · LazyVideo · Rune · SectionHeading
  pages/         Landing · GamePage
  sections/      Hero · Story · Features · Characters · Maps · Trailer · Gallery
                  SystemRequirements · Download · home/HeroBackdrop

Work in order: scaffold and theme, then data and pipeline, then shell, then portal, then sections,
then media/audio, then tests. After each stage, tell me what changed and what to look at.
When you finish, report: the build output with chunk sizes, the ESLint result, the Playwright
results, and any place where you had to ask me about the artwork.
</output_format>

<final_check>
Before finishing, verify against the requirements above and confirm every item:
- npm run build succeeds with no warnings; npm run lint is clean.
- Playwright, against the preview build, passes: no console errors, exceptions or failed requests
  on all four routes; correct data-theme, --accent and display font per route; rendered section
  order equals configuration for each world; no boss-style section is rendered that no world
  lists; Valorant character cards and the map plate render no DOM captions; no iframe exists
  before the trailer is clicked and a youtube-nocookie iframe exists after; gallery clips are not
  mounted before scrolling; the lightbox opens, advances, wraps backwards and closes on Escape,
  and returns focus; external storefront links carry rel="noopener"; window.__lenis and the custom
  cursor are absent under prefers-reduced-motion and present at full motion; an unknown route

```

renders the portal; a nav click changes theme without a reload; at 390px there is no horizontal overflow on any route, the portrait plate shows three tap bands, and Motion and Fullscreen are reachable in the menu sheet; cumulative layout shift stays below 0.1 on every route.

- Re-running npm run assets twice in a row converts nothing the second time and leaves manifest.json byte-identical.
- No hard-coded /assets path outside assets.js; no duplicate object keys; no unreferenced component left in src/.
- The README matches what the code actually does.

</final_check>